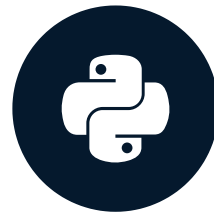


Model metrics and adjustments

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)



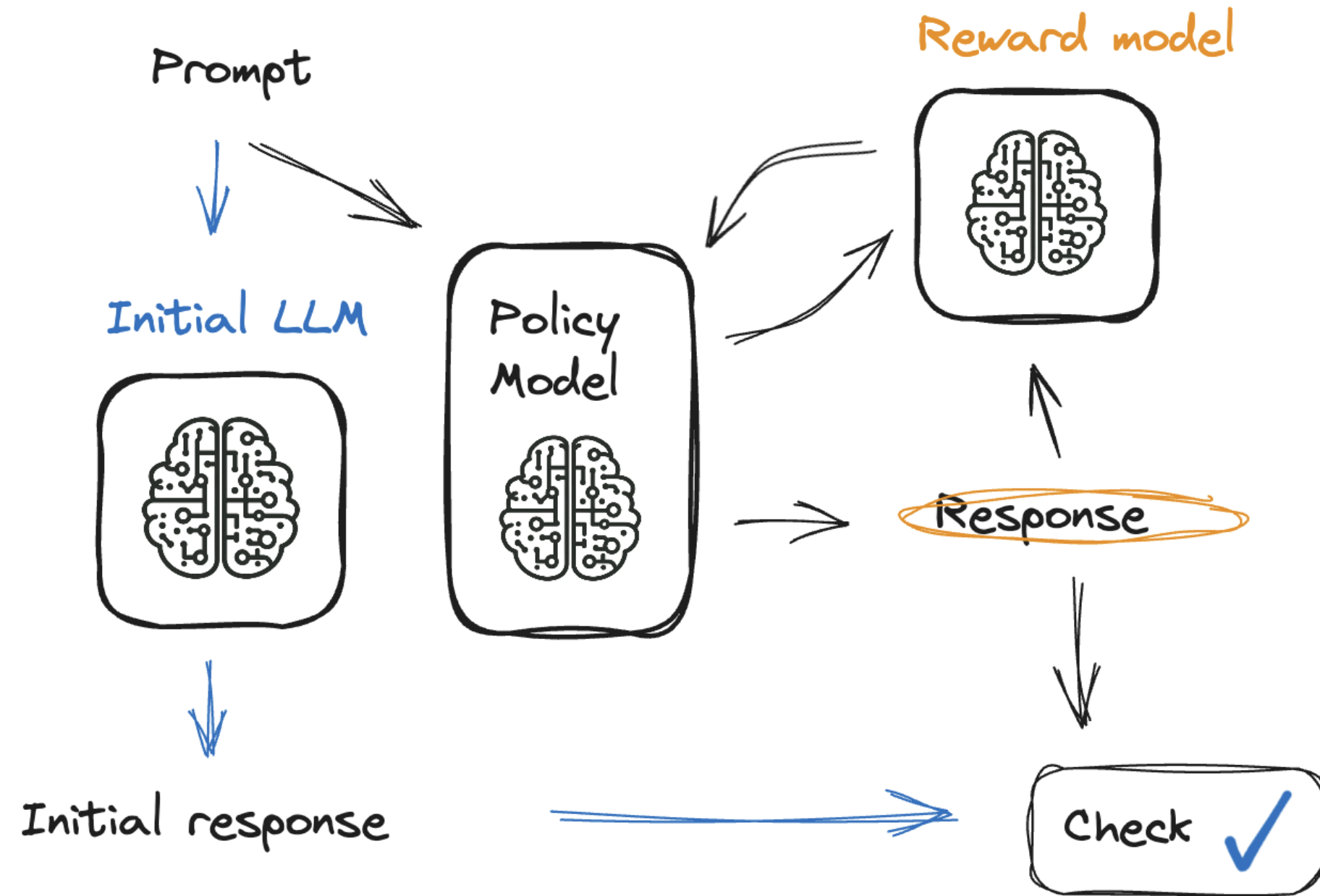
Mina Parham
AI Engineer

Why use a reference model?

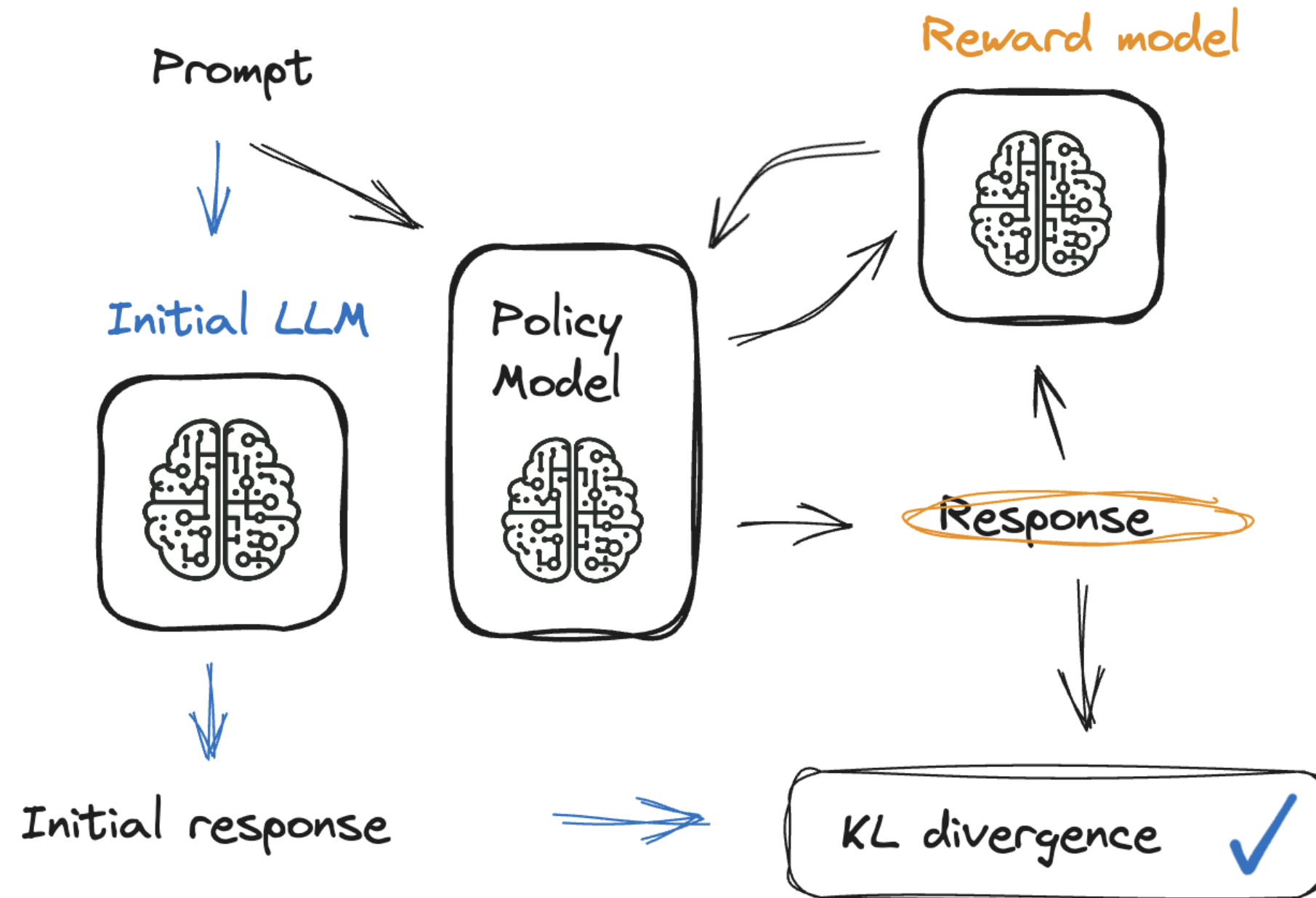
- Meaningless outputs



Checking model output



Solution: KL divergence



Solution: KL divergence

- **Penalty** is added to the reward model
- Penalty **redirects** the model if it produces unrelated outputs
- **KL divergence** compares current and reward model



- Between 0 and 10, and **never negative**

Adjusting parameters

```
generation_kwargs = {  
    "min_length": -1, # don't ignore the EOS token  
    "top_k": 0.0, # no top-k sampling  
    "top_p": 1.0,  
    "do_sample": True,  
    "pad_token_id": tokenizer.eos_token_id,  
    "max_new_tokens": 32}
```

- Parameters are passed to the policy model

Checking the reward model

- Checking the reward model
- Checking output (reward)

```
reward_model_results.head()
```

ID	Comment	Sentiment	Reward
1	This event was lit! So much fun!	Positive	0.9
2	Terrible experience, never attending again.	Negative	-0.8
3	It was okay, nothing extraordinary.	Neutral	0.2
4	The event was poorly organized and chaotic.	Negative	-0.85
5	Had an amazing time with great people!	Positive	0.95

Checking the reward model

- □ □ Check extreme cases

```
extreme_positive = reward_model_results[reward_model_results['Reward'] >= 0.9]  
extreme_negative = reward_model_results[reward_model_results['Reward'] <= -0.8]
```

- □ Ensure balanced dataset

```
sentiment_distribution = reward_model_results['Sentiment'].value_counts()
```

- □ Normalize the reward model

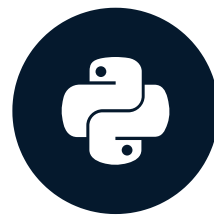
```
from sklearn.preprocessing import MinMaxScaler  
scaler = MinMaxScaler(feature_range=(-1, 1))  
scaler.fit_transform(reward_model_results[['Reward']])
```

Let's practice!

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Incorporating diverse feedback sources

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)



Mina Parham
AI Engineer

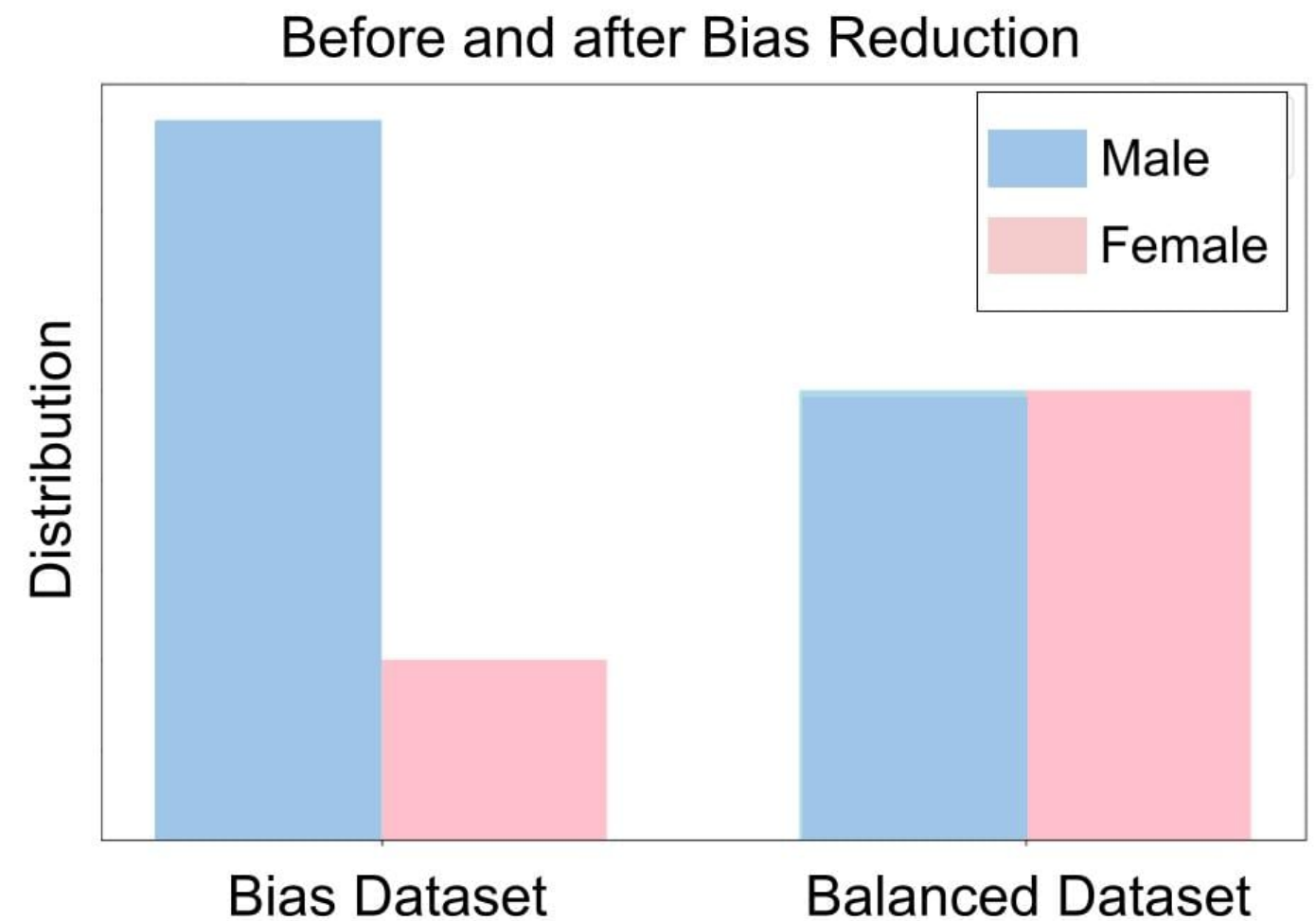
Improved model generalization

- Represent different viewpoints and contexts
- Generalizes preferences and values



Reduced bias

- Mitigates individual biases
- Creates a more balanced and fair model output



Better alignment with human values

- Complex human preferences
- Cultures and backgrounds represented



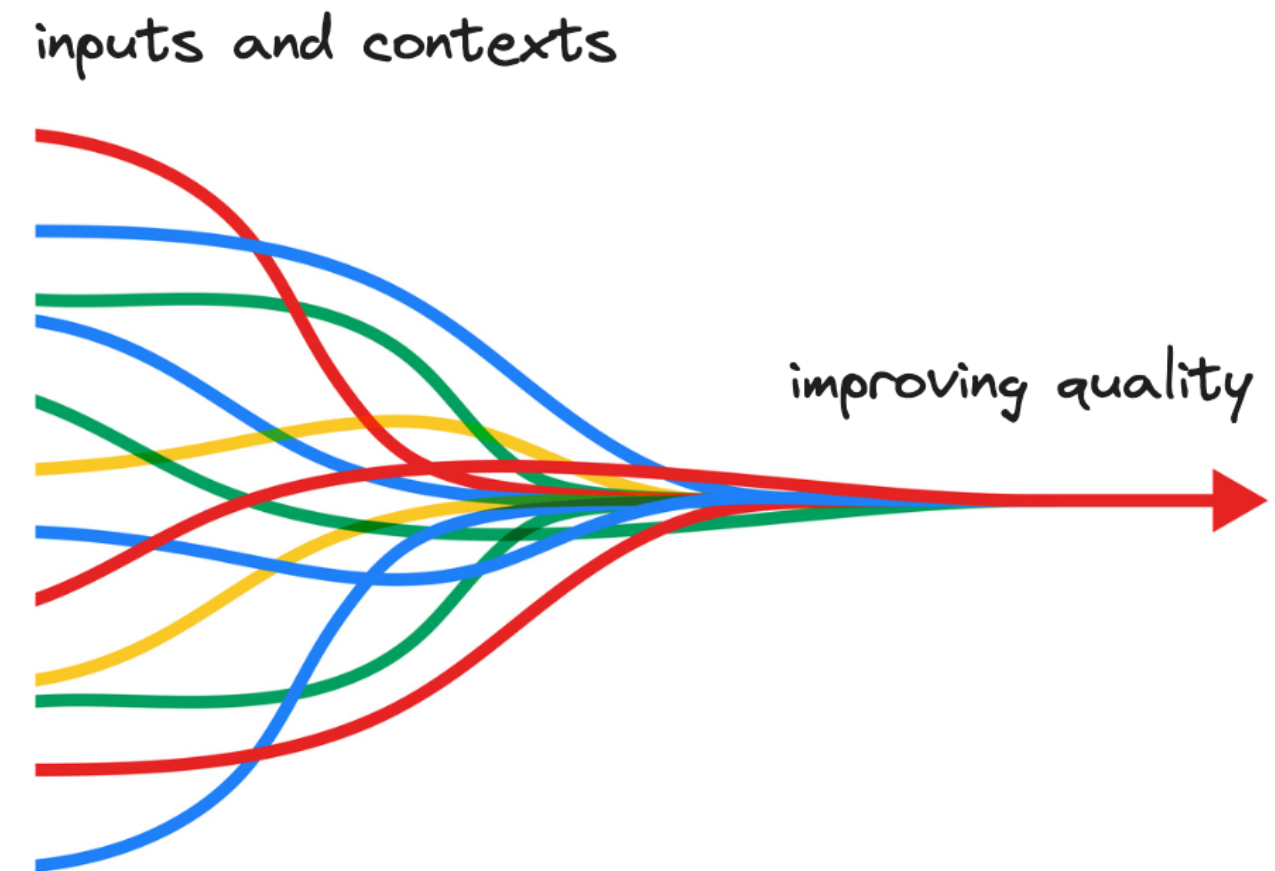
Enhanced adaptability

- Model responds to wider range of user needs and preferences
- Represent different viewpoints



Increased robustness

- Resilient to different types of inputs
- Improving its performance



Integrating preference data from multiple sources

Preference data `preference_df` with sources `'Journalist'`, `'Social Media Influencer'`, and `'Marketing Professional'`:

	source	id	chosen	rejected
0	Journalist	1	Global climate summit reaches landmark agreement	World leaders make promise about weather
1	Journalist	2	Tech giant unveils groundbreaking AI chip	New computer part is super fast
2	Journalist	3	Supreme Court rules on controversial privacy case	Judges decide on big secret case
3	Social Media Influencer	1	World leaders make promise about weather	Global climate summit reaches landmark agreement
4	Social Media Influencer	2	Tech giant unveils groundbreaking AI chip	New computer part is super fast
5	Social Media Influencer	3	Judges decide on big secret case	Supreme Court rules on controversial privacy case
6	Marketing Professional	1	Global climate summit reaches landmark agreement	World leaders make promise about weather
7	Marketing Professional	2	New computer part is super fast	Tech giant unveils groundbreaking AI chip
8	Marketing Professional	3	Supreme Court rules on controversial privacy case	Judges decide on big secret case

Majority voting

This sample data could easily be integrated by grouping by 'id':

```
df_majority = preference_df.groupby(['id']).apply(majority_vote)
```

Then, using majority voting:

```
from collections import Counter

def majority_vote(df):
    votes = Counter(zip(df['chosen'], df['rejected']))
    return max(votes, key=votes.get)
```

Unreliable preference data sources

Preference data `preference_df2` with the same three experts:

	source	id	chosen	rejected
0	Journalist	1	Study shows climate change accelerating faster than predicted	Weather patterns changing, scientists unsure why
1	Journalist	2	New cancer treatment shows promising results in clinical trials	Doctors test new medicine for serious illness
2	Journalist	3	Tech company launches innovative privacy protection software	Company creates program to keep information safe
3	Journalist	4	Global economy faces challenges amid rising inflation rates	Money problems worry people around the world
4	Journalist	5	Renewable energy investments reach record high in 2024	More money spent on clean energy than ever before
5	Social Media Influencer	1	Study shows climate change accelerating faster than predicted	Weather patterns changing, scientists unsure why
6	Social Media Influencer	2	New cancer treatment shows promising results in clinical trials	Doctors test new medicine for serious illness
7	Social Media Influencer	3	Tech company launches innovative privacy protection software	Company creates program to keep information safe
8	Social Media Influencer	4	Global economy faces challenges amid rising inflation rates	Money problems worry people around the world
9	Social Media Influencer	5	Renewable energy investments reach record high in 2024	More money spent on clean energy than ever before
10	Marketing Professional	1	Weather patterns changing, scientists unsure why	Study shows climate change accelerating faster than predicted
11	Marketing Professional	2	Doctors test new medicine for serious illness	New cancer treatment shows promising results in clinical trials
12	Marketing Professional	3	Company creates program to keep information safe	Tech company launches innovative privacy protection software
13	Marketing Professional	4	Money problems worry people around the world	Global economy faces challenges amid rising inflation rates
14	Marketing Professional	5	More money spent on clean energy than ever before	Renewable energy investments reach record high in 2024

Unreliable preference data sources

- Iterating over the rows of `preference_df2` to identify unreliable sources:

```
df_majority = preference_df2.groupby('id').apply(majority_vote)
disagreements = {source: 0 for source in preference_df2['source'].unique()}

for _, row in preference_df2.iterrows():
    if (row['chosen'], row['rejected']) != df_majority[row['id']]:
        disagreements[row['source']] += 1

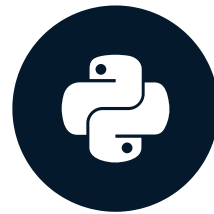
detect_unreliable_source = max(disagreements, key=disagreements.get)
```

Let's practice!

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Evaluating RLHF models

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)



Mina Parham
AI Engineer

Automation metrics

- **Classification task:** Accuracy, F1 score

```
classification_results.head(3)
```

ID	Feedback_Text	True_Category	Predicted_Category
1	"Arrived on time and works great."	Positive	Positive
2	"I had issues with customer service."	Negative	Neutral
3	"The website is easy to navigate."	Positive	Positive

Automation metrics

- Text generation, summarization: ROUGE, BLEU

```
text_generation.head(3)
```

ID	Prompt	True_Completion	Pred_Completion
1	"Customer service"	"can help you."	"will assist."
2	"To get a refund,"	"contact us."	"reach out."
3	"Support team is"	"here 24/7."	"available 24/7."

Automation metrics

Reference statement:

- **RLHF improves model alignment with human values.**

Statement to compare:

- **RLHF aligns models with human values.**

ROUGE score: 0.83

Artifact curves

```
config = PP0Config(  
    model_name="lvwerra/gpt2-imdb", learning_rate=1.41e-5, log_with="wandb")
```

```
import wandb  
wandb.init()
```



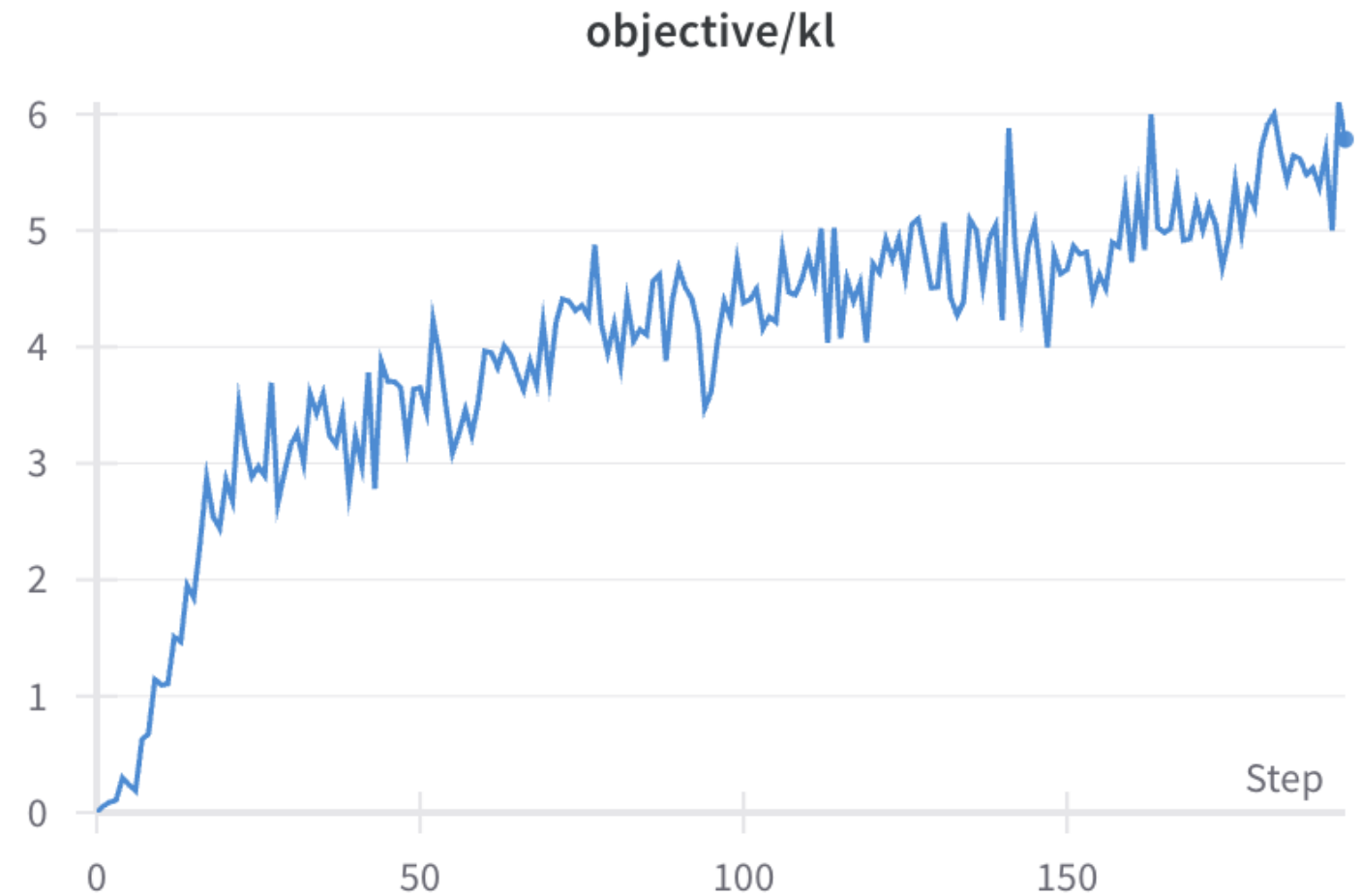
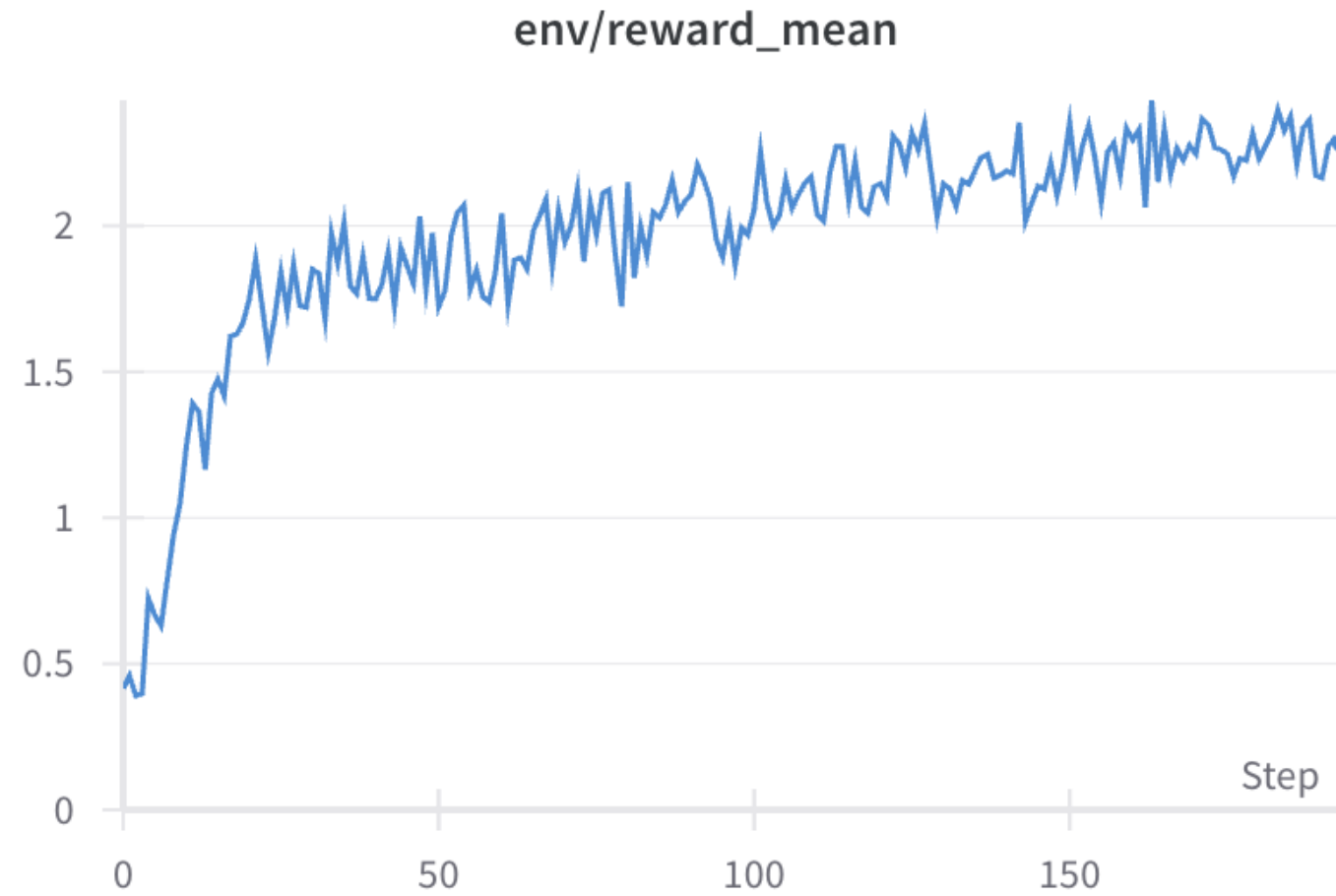
A terminal window with a light gray background. The top bar contains icons for navigation and settings. The terminal shows the execution of two Python commands: `import wandb` and `wandb.init()`. The output includes several informational messages from the `wandb` package, such as using the SDK backend, logging into wandb.ai, and providing a link to the API key. A text input field is visible at the bottom of the terminal output.

```
1 import wandb  
2 wandb.init()  
  
... wandb: Using wandb-core as the SDK backend. Please refer to https://wandb.me/wandb-core for more information.  
wandb: Logging into wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)  
wandb: You can find your API key in your browser here: https://wandb.ai/authorize  
wandb: Paste an API key from your profile and hit enter, or press ctrl+c to quit:  

```

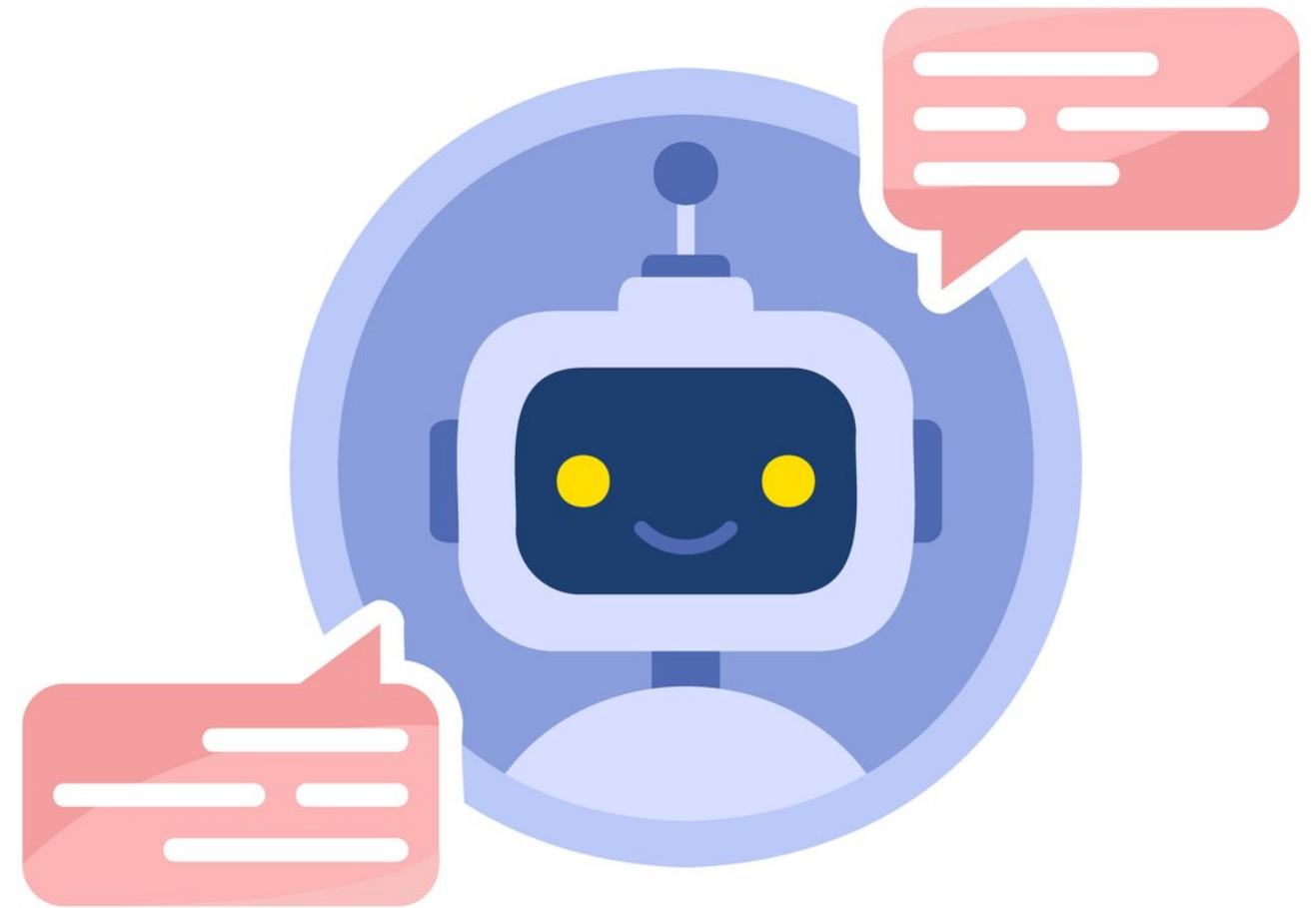
Artifact curves

- Reward increases as the model learns.
- The KL curve should increase gradually.



Human centered evaluation

- Human evaluation: subjective judgments or a deep understanding of context
- Models evaluation: scalability and consistency

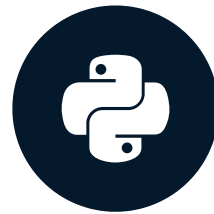


Let's practice!

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

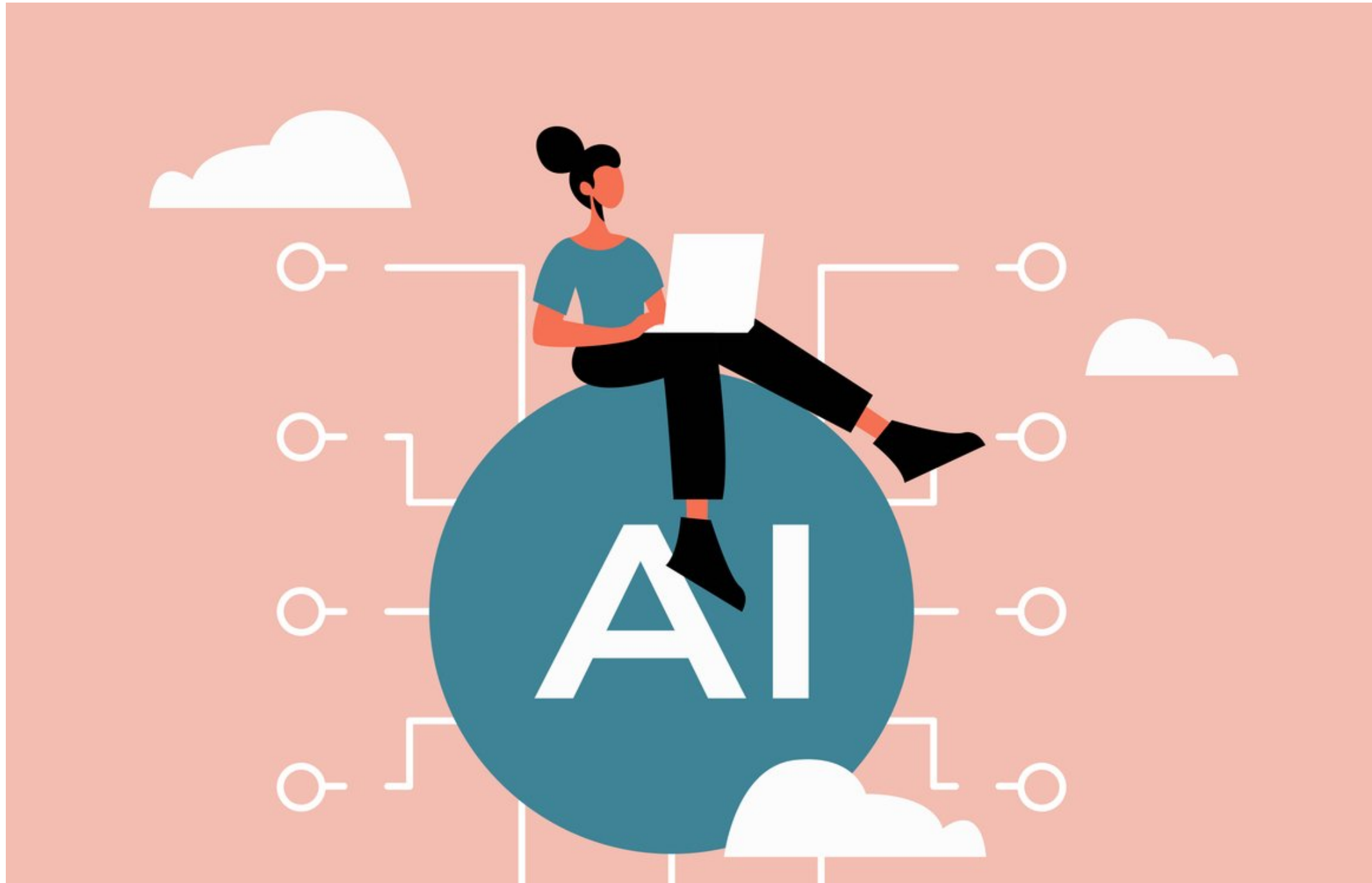
Wrapping up your RLHF journey

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

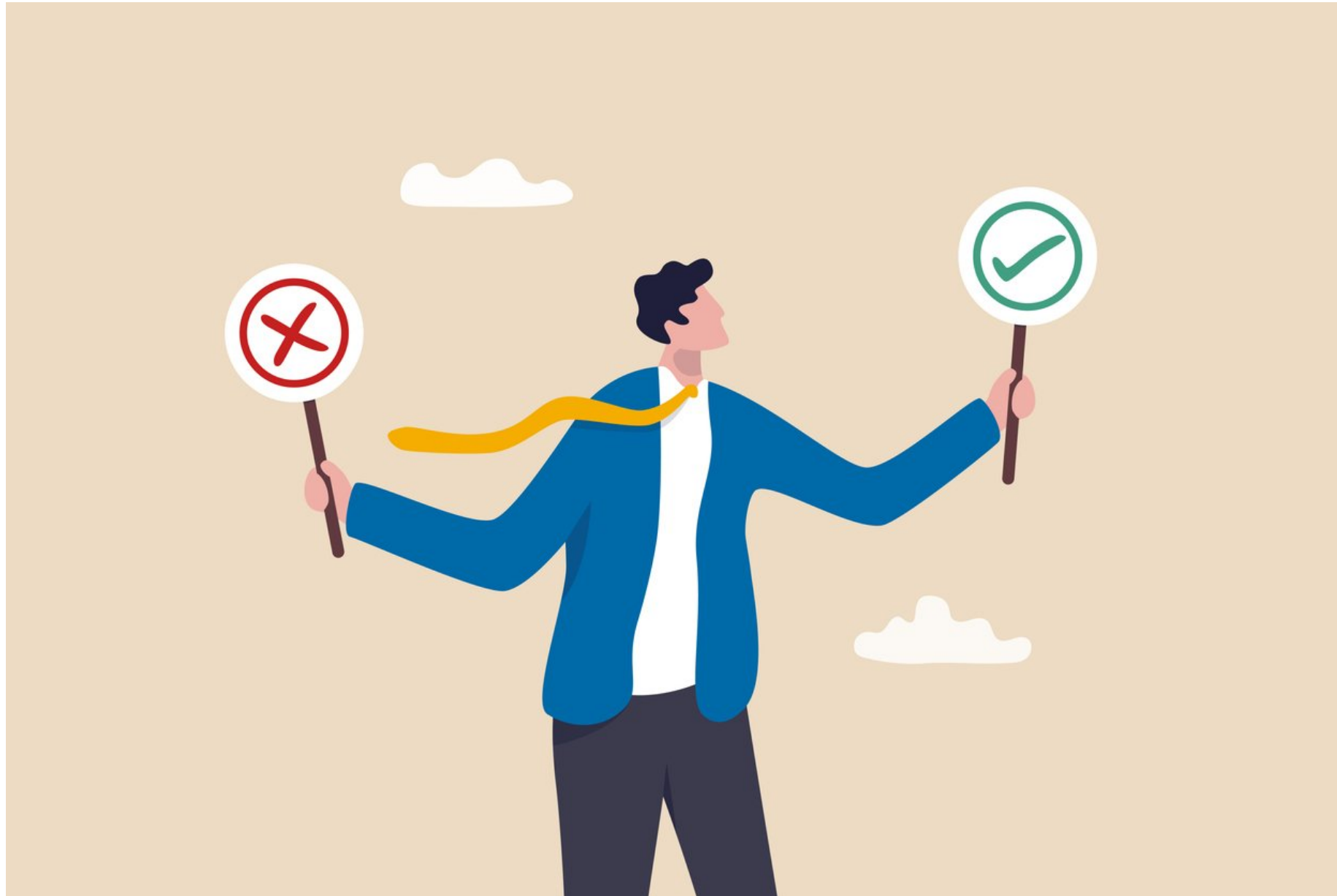


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AI Engineer

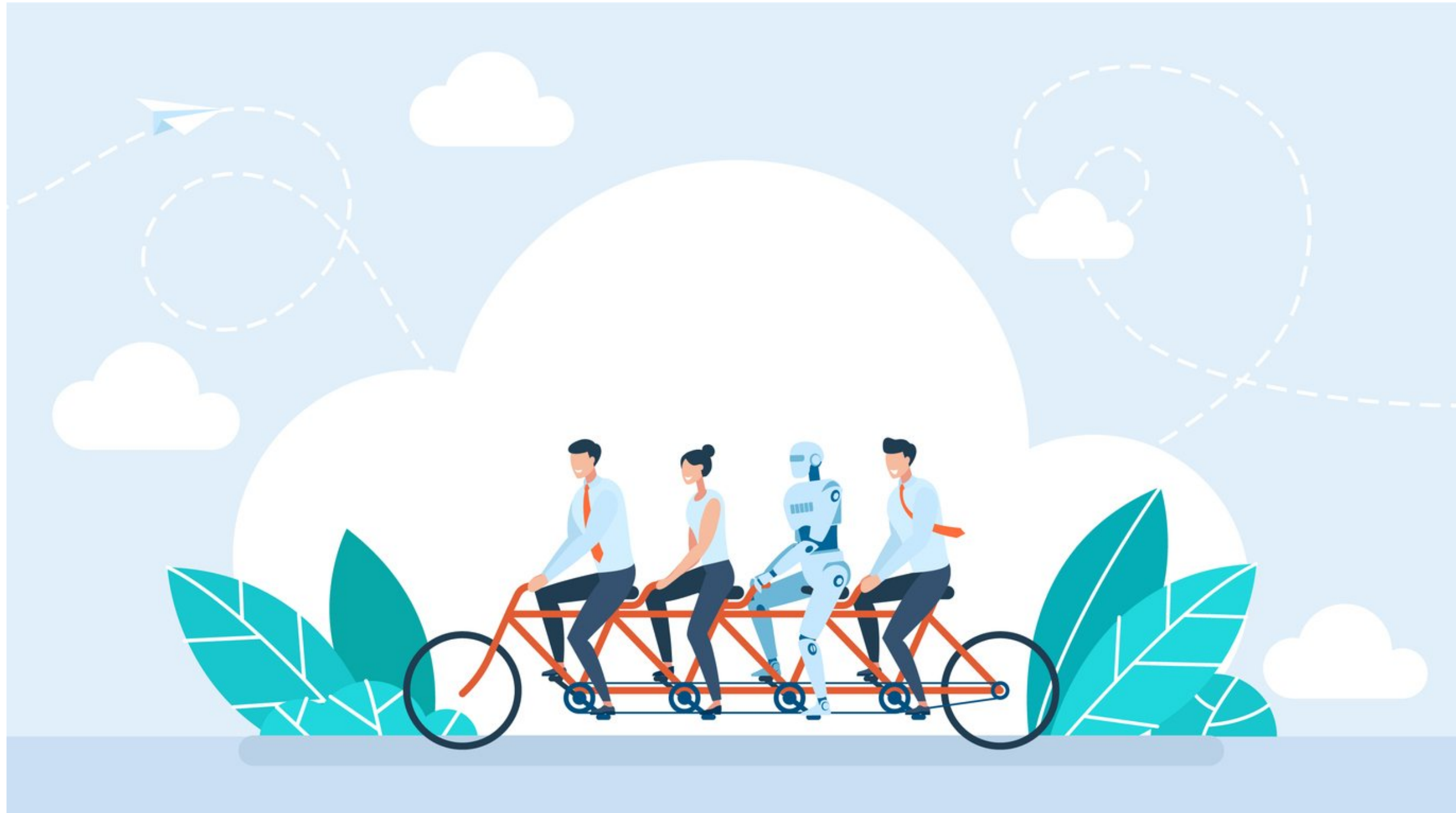
Starting the journey with foundational concepts



Gathering high-quality feedback



Reward models and human feedback in the loop



Metrics and evaluation



Congratulations!

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)